



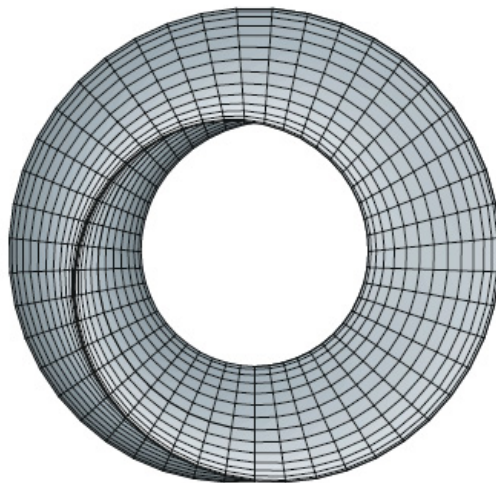
UNIVERSITY OF ZIMBABWE

HMTHCS111 CALCULUS 2

Calculus of Several Variables

Author:
G. NHAWU

Department:
MATHEMATICS AND
COMPUTATIONAL SCIENCES



August 28, 2026

Contents

5	Vector-Valued Functions	2
5.1	Plane Curves and Parametric Equations	2
5.2	Limits and Continuity	3
5.3	Definition of the Limit of a Vector-Valued Function	3
5.4	Differentiation and Integration of Vector-Valued Functions	5
5.4.1	Differentiation of Vector-Valued Functions	5
5.5	Properties of the Derivative	7
5.6	Integration of Vector-Valued Functions	8
5.7	Velocity and Acceleration	9
5.8	Tangent Vectors and Normal Vectors	11
5.8.1	Definition of Unit Tangent Vector	11
5.9	Tutorial Questions	12

Chapter 5

Vector-Valued Functions

If you think dogs can't count, try putting three dog biscuits in your pocket and then giving Fido only two of them.

—Phil Pastoret

This chapter introduces the concept of vector-valued functions. Vector-valued functions can be used to study curves in the plane and in space. These functions can also be used to study the motion of an object along a curve.

5.1 Plane Curves and Parametric Equations

Until now, we have been representing a graph by a single equation involving two variables. Consider the path followed by an object that is propelled into the air at an angle of 45° . If the initial velocity of the object is 48 meters per second, the object travels the parabolic path given by

$$y = -\frac{x^2}{72} + x, \quad \text{Rectangular form.}$$

However, this equation does not tell the whole story. Although it does tell you where the object has been, it doesn't tell you when the object was at a given point (x, y) . To determine this time, you introduce a third variable t , called a **parameter**. By writing both x and y as functions of t , you obtain the **parametric equations**

$$x = 24\sqrt{2}t \quad \text{and} \quad y = -16t^2 + 24\sqrt{2}t \quad \text{Parametric equations.}$$

From this set of equations, you can determine that at time $t = 0$, the object is at the point $(0, 0)$. Similarly, at time $t = 1$, the object is at the point $(24\sqrt{2}, 24\sqrt{2} - 16)$, and so on.

Theorem 5.1.1. If f and g are continuous functions of t on an interval I , then the equations

$$x = f(t) \quad \text{and} \quad y = g(t)$$

are called **parametric equations** and t is called a **parameter**.

A **space curve** is the set of all ordered triples $(f(t), g(t), h(t))$ together with their defining parametric equations

$$x = f(t), \quad y = g(t) \quad \text{and} \quad z = h(t)$$

where f, g and h are continuous functions of t on the interval I .

Theorem 5.1.2. A function of the form

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} \quad \text{Plane}$$

or

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k} \quad \text{Space}$$

is a **vector-valued function**, where the **component functions** f, g and h are real-valued functions of the parameter t . Vector-valued functions are sometimes denoted as $\mathbf{r}(t) = (f(t), g(t))$ or $\mathbf{r}(t) = (f(t), g(t), h(t))$.

Vector-valued functions serve dual roles in the representation of curves. By letting the parameter t represent time, we can use a vector-valued function to represent *motion* along a curve. Or, in the more general case, we can use a vector-valued function to *trace the graph* of a curve. Unless stated otherwise, the **domain** of a vector-valued function $\mathbf{r}$ is considered to be the intersection of the domains of the component functions f, g and h . For example, the domain of $\mathbf{r}(t) = \ln t \mathbf{i} + \sqrt{1-t} \mathbf{j} + t \mathbf{k}$ is the interval $(0, 1]$.

5.2 Limits and Continuity

Many techniques and definitions used in calculus of real-valued functions can be applied to vector-valued functions. For example, we can add and subtract vector-valued functions, multiply a vector-valued function with a scalar, take the limit of a vector-valued function, differentiate a vector-valued function, and so on.

$$\begin{aligned} \mathbf{r}_1(t) + \mathbf{r}_2(t) &= [f_1(t)\mathbf{i} + g_1(t)\mathbf{j}] + [f_2(t)\mathbf{i} + g_2(t)\mathbf{j}] \\ &= [f_1(t) + f_2(t)]\mathbf{i} + [g_1(t) + g_2(t)]\mathbf{j}. \end{aligned}$$

5.3 Definition of the Limit of a Vector-Valued Function

If $\mathbf{r}$ is a vector-valued function such that $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}$, then

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \left[\lim_{t \rightarrow a} f(t) \right] \mathbf{i} + \left[\lim_{t \rightarrow a} g(t) \right] \mathbf{j} \quad \text{Plane}$$

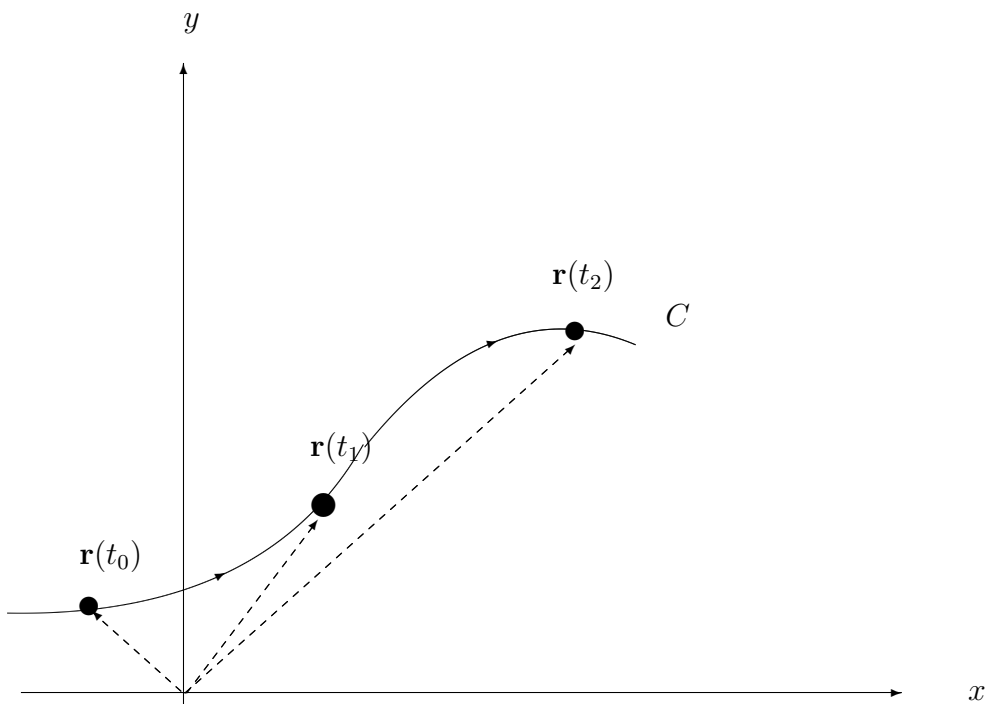


Figure 5.1: Curve in plane

provided f and g have limits as $t \rightarrow a$.

If $\mathbf{r}$ is a vector-valued function such that $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, then

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \left[\lim_{t \rightarrow a} f(t) \right] \mathbf{i} + \left[\lim_{t \rightarrow a} g(t) \right] \mathbf{j} + \left[\lim_{t \rightarrow a} h(t) \right] \mathbf{k} \quad \text{Space}$$

provided f, g and h have limits as $t \rightarrow a$.

If $\mathbf{r}(t)$ approaches the vector $\mathbf{L}$ as $t \rightarrow a$, the length of the vector $\mathbf{r}(t) - \mathbf{L}$ approaches 0, that is

$$|\mathbf{r}(t) - \mathbf{L}| \rightarrow 0 \quad \text{as } t \rightarrow a.$$

The next theorem extends the notion of continuity of vector-valued functions.

Theorem 5.3.1. *A vector-valued function $\mathbf{r}$ is **continuous at a point** given by $t = a$ if the limit of $\mathbf{r}(t)$ exists as $t \rightarrow a$ and*

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \mathbf{r}(a).$$

A vector-valued function $\mathbf{r}$ is **continuous on an interval** I if it is continuous at every point in the interval. It follows that a vector-valued function is continuous at $t = a$ if and only if each of its component functions is continuous at $t = a$.

Example 5.3.1. *Discuss the continuity of the vector-valued function given by*

$$\mathbf{r}(t) = t\mathbf{i} + a\mathbf{j} + (a^2 - t^2)\mathbf{k}$$

at $t = 0$.

Solution: As t approaches 0, the limit is

$$\begin{aligned}\lim_{t \rightarrow 0} \mathbf{r}(t) &= \left[\lim_{t \rightarrow 0} t \right] \mathbf{i} + \left[\lim_{t \rightarrow 0} a \right] \mathbf{j} + \left[\lim_{t \rightarrow 0} (a^2 - t^2) \right] \mathbf{k} \\ &= 0\mathbf{i} + a\mathbf{j} + a^2\mathbf{k} \\ &= a\mathbf{j} + a^2\mathbf{k}.\end{aligned}$$

Because

$$\begin{aligned}\mathbf{r}(0) &= (0)\mathbf{i} + (a)\mathbf{j} + (a^2)\mathbf{k} \\ &= a\mathbf{j} + a^2\mathbf{k}.\end{aligned}$$

We can conclude that $\mathbf{r}$ is continuous at $t = 0$.

5.4 Differentiation and Integration of Vector-Valued Functions

5.4.1 Differentiation of Vector-Valued Functions

The definition of the derivative of a vector-valued function parallels the definition given for real-valued functions.

Theorem 5.4.1. *The **derivative of a vector-valued function** $\mathbf{r}$ is defined by*

$$\mathbf{r}'(t) = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t}$$

for all t for which the limit exists. If $\mathbf{r}'(t)$ exists, then $\mathbf{r}$ is **differentiable at t** . In addition to $\mathbf{r}'(t)$, other notations for the derivative of a vector-valued function are

$$D_t[\mathbf{r}(t)], \quad \frac{d}{dt}[\mathbf{r}(t)] \quad \text{and} \quad \frac{d\mathbf{r}}{dt}.$$

Differentiation of vector-valued functions can be done on a *component-by-component basis*. To see why this is true, consider the function given by

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}.$$

Applying the definition of the derivative produces the following

$$\begin{aligned}
\mathbf{r}'(t) &= \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t + \Delta t) - \mathbf{r}(t)}{\Delta t} \\
&= \lim_{\Delta t \rightarrow 0} \frac{f(t + \Delta t)\mathbf{i} + g(t + \Delta t)\mathbf{j} - f(t)\mathbf{i} - g(t)\mathbf{j}}{\Delta t} \\
&= \lim_{\Delta t \rightarrow 0} \left\{ \left[\frac{f(t + \Delta t) - f(t)}{\Delta t} \right] \mathbf{i} + \left[\frac{g(t + \Delta t) - g(t)}{\Delta t} \right] \mathbf{j} \right\} \\
&= \left\{ \lim_{\Delta t \rightarrow 0} \left[\frac{f(t + \Delta t) - f(t)}{\Delta t} \right] \mathbf{i} \right\} + \left\{ \lim_{\Delta t \rightarrow 0} \left[\frac{g(t + \Delta t) - g(t)}{\Delta t} \right] \mathbf{j} \right\} \\
&= f'(t)\mathbf{i} + g'(t)\mathbf{j}.
\end{aligned}$$

Note that the derivative of the vector-valued function $\mathbf{r}$ is itself a vector-valued function. $\mathbf{r}'(t)$ is a vector tangent to the curve given by $\mathbf{r}(t)$ and pointing in the direction of increasing t -values.

Theorem 5.4.2. If $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}$, where f and g are differentiable functions of t , then

$$\mathbf{r}'(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j}.$$

If $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, where f, g and h are differentiable functions of t , then

$$\mathbf{r}'(t) = f'(t)\mathbf{i} + g'(t)\mathbf{j} + h'(t)\mathbf{k}.$$

Example 5.4.1. For the vector-valued function given by $\mathbf{r}(t) = t\mathbf{i} + (t^2 + 2)\mathbf{j}$, find $\mathbf{r}'(t)$.

Solution: $\mathbf{r}'(t) = \mathbf{i} + 2t\mathbf{j}$.

Higher-order derivatives of vector-valued functions are obtained by successive differentiation of each component function.

Example 5.4.2. For the vector-valued function given by $\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + 2t\mathbf{k}$, find each of the following (i) $\mathbf{r}'(t)$ (ii) $\mathbf{r}''(t)$ (iii) $\mathbf{r}'(t) \cdot \mathbf{r}''(t)$ (iv) $\mathbf{r}'(t) \times \mathbf{r}''(t)$.

Solution: (i) $\mathbf{r}'(t) = -\sin t\mathbf{i} + \cos t\mathbf{j} + 2\mathbf{k}$.

(ii) $\mathbf{r}''(t) = -\cos t\mathbf{i} - \sin t\mathbf{j} + 0\mathbf{k} = -\cos t\mathbf{i} - \sin t\mathbf{j}$.

(iii) $\mathbf{r}'(t) \cdot \mathbf{r}''(t) = \sin t \cos t - \sin t \cos t = 0$.

(iv) $\mathbf{r}'(t) \times \mathbf{r}''(t) = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -\sin t & \cos t & 2 \\ -\cos t & -\sin t & 0 \end{vmatrix} = \begin{vmatrix} \cos t & 2 \\ -\sin t & 0 \end{vmatrix} \mathbf{i} - \begin{vmatrix} -\sin t & 2 \\ -\cos t & 0 \end{vmatrix} \mathbf{j} + \begin{vmatrix} -\sin t & \cos t \\ -\cos t & -\sin t \end{vmatrix} \mathbf{k}$
 $= 2 \sin t\mathbf{i} - 2 \cos t\mathbf{j} + \mathbf{k}.$

The parametrization of the curve represented by the vector-valued function

$$\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$$

is **smooth on an open interval I** if f', g' and h' are continuous on I and $\mathbf{r}'(t) \neq 0$ for any value of t in the interval I .

Example 5.4.3. Find the intervals on which the epicycloid C given by

$$\mathbf{r}(t) = (5 \cos t - \cos 5t)\mathbf{i} + (5 \sin t - \sin 5t)\mathbf{j}, \quad 0 \leq t \leq 2\pi$$

is smooth.

Solution: The derivative of $\mathbf{r}$ is

$$\mathbf{r}'(t) = (-5 \sin t + 5 \sin 5t)\mathbf{i} + (5 \cos t - 5 \cos 5t)\mathbf{j}.$$

In the interval $[0, 2\pi]$, the only values of t for which

$$\mathbf{r}'(t) = 0\mathbf{i} + 0\mathbf{j}$$

are $t = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$. Therefore, we conclude that C is smooth in the intervals

$$\left(0, \frac{\pi}{2}\right), \left(\frac{\pi}{2}, \pi\right), \left(\pi, \frac{3\pi}{2}\right) \text{ and } \left(\frac{3\pi}{2}, 2\pi\right).$$

5.5 Properties of the Derivative

Let $\mathbf{r}$ and $\mathbf{u}$ be differentiable vector-valued functions of t , let w be a differentiable real-valued function of t , and let c be a scalar.

1. $D_t[\mathbf{c}\mathbf{r}(t)] = \mathbf{c}\mathbf{r}'(t)$.
2. $D_t[\mathbf{r}(t) \pm \mathbf{u}(t)] = \mathbf{r}'(t) \pm \mathbf{u}'(t)$.
3. $D_t[w(t)\mathbf{r}(t)] = w(t)\mathbf{r}'(t) + w'(t)\mathbf{r}(t)$.
4. $D_t[\mathbf{r}(t) \cdot \mathbf{u}(t)] = \mathbf{r}(t) \cdot \mathbf{u}'(t) + \mathbf{r}'(t) \cdot \mathbf{u}(t)$.
5. $D_t[\mathbf{r}(t) \times \mathbf{u}(t)] = \mathbf{r}(t) \times \mathbf{u}'(t) + \mathbf{r}'(t) \times \mathbf{u}(t)$.
6. $D_t[\mathbf{r}(w(t))] = \mathbf{r}'(w(t))w'(t)$.
7. If $\mathbf{r}(t) \cdot \mathbf{r}(t) = c$, then $\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$.

To prove property 4, let $\mathbf{r}(t) = f_1(t)\mathbf{i} + g_1(t)\mathbf{j}$ and $\mathbf{u}(t) = f_2(t)\mathbf{i} + g_2(t)\mathbf{j}$, where f_1, f_2, g_1 and g_2 are differentiable functions of t . Then

$$\mathbf{r}(t) \cdot \mathbf{u}(t) = f_1(t)f_2(t) + g_1(t)g_2(t)$$

and it follows that

$$\begin{aligned} D_t[\mathbf{r}(t) \cdot \mathbf{u}(t)] &= f_1(t)f_2'(t) + f_1'(t)f_2(t) + g_1(t)g_2'(t) + g_1'(t)g_2(t) \\ &= [f_1(t)f_2'(t) + g_1(t)g_2'(t)] + [f_1'(t)f_2(t) + g_1'(t)g_2(t)] \\ &= \mathbf{r}(t) \cdot \mathbf{u}'(t) + \mathbf{r}'(t) \cdot \mathbf{u}(t). \end{aligned}$$

Example 5.5.1. For the vector-valued function given by

$$\mathbf{r}(t) = \frac{1}{t}\mathbf{i} - \mathbf{j} + \ln t\mathbf{k} \quad \text{and} \quad \mathbf{u}(t) = t^2\mathbf{i} - 2t\mathbf{j} + \mathbf{k}$$

find (i) $D_t[\mathbf{r}(t) \cdot \mathbf{u}(t)]$ (ii) $D_t[\mathbf{u}(t) \times \mathbf{u}'(t)]$.

Solution: (i) Because $\mathbf{r}'(t) = -\frac{1}{t^2}\mathbf{i} + \frac{1}{t}\mathbf{k}$ and $\mathbf{u}'(t) = 2t\mathbf{i} - 2\mathbf{j}$, we have

$$\begin{aligned} D_t[\mathbf{r}(t) \cdot \mathbf{u}(t)] &= \mathbf{r}(t) \cdot \mathbf{u}'(t) + \mathbf{r}'(t) \cdot \mathbf{u}(t) \\ &= \left(\frac{1}{t}\mathbf{i} - \mathbf{j} + \ln t\mathbf{k} \right) \cdot (2t\mathbf{i} - 2\mathbf{j}) + \left(-\frac{1}{t^2}\mathbf{i} + \frac{1}{t}\mathbf{k} \right) \cdot (t^2\mathbf{i} - 2t\mathbf{j} + \mathbf{k}) \\ &= 2 + 2 + (-1) + \frac{1}{t} \\ &= 3 + \frac{1}{t}. \end{aligned}$$

(ii) Because $\mathbf{u}'(t) = 2t\mathbf{i} - 2\mathbf{j}$ and $\mathbf{u}''(t) = 2\mathbf{i}$, we have

$$\begin{aligned} D_t[\mathbf{u}(t) \times \mathbf{u}'(t)] &= [\mathbf{u}(t) \times \mathbf{u}''(t)] + [\mathbf{u}'(t) \times \mathbf{u}'(t)] \\ &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ t^2 & -2t & 1 \\ 2 & 0 & 0 \end{vmatrix} + 0 \\ &= 2\mathbf{j} + 4t\mathbf{k}. \end{aligned}$$

5.6 Integration of Vector-Valued Functions

If $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j}$, where f and g are continuous on $[a, b]$, then the **indefinite integral (anti-derivative)** of $\mathbf{r}$ is

$$\int \mathbf{r}(t) dt = \left[\int f(t) dt \right] \mathbf{i} + \left[\int g(t) dt \right] \mathbf{j}$$

and its **definite integral** over the interval $a \leq t \leq b$ is

$$\int_a^b \mathbf{r}(t) dt = \left[\int_a^b f(t) dt \right] \mathbf{i} + \left[\int_a^b g(t) dt \right] \mathbf{j}.$$

If $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$, where f, g and h are continuous on $[a, b]$, then the **indefinite integral (anti-derivative)** of $\mathbf{r}$ is

$$\int \mathbf{r}(t) dt = \left[\int f(t) dt \right] \mathbf{i} + \left[\int g(t) dt \right] \mathbf{j} + \left[\int h(t) dt \right] \mathbf{k}$$

and its **definite integral** over the interval $a \leq t \leq b$ is

$$\int_a^b \mathbf{r}(t) dt = \left[\int_a^b f(t) dt \right] \mathbf{i} + \left[\int_a^b g(t) dt \right] \mathbf{j} + \left[\int_a^b h(t) dt \right] \mathbf{k}.$$

The anti-derivative of a vector-valued function is a family of vector-valued functions all differing by a constant vector $\mathbf{C}$.

Example 5.6.1. Find the indefinite integral

$$\int (t\mathbf{i} + 3\mathbf{j}) dt.$$

Solution: Integrating on a component-by-component basis produces

$$\int (t\mathbf{i} + 3\mathbf{j}) dt = \frac{t^2}{2}\mathbf{i} + 3t\mathbf{j} + \mathbf{C}.$$

Example 5.6.2. Evaluate the integral

$$\int_0^1 \mathbf{r}(t) dt = \int_0^1 \left(\sqrt[3]{t}\mathbf{i} + \frac{1}{t+1}\mathbf{j} + e^{-t}\mathbf{k} \right) dt.$$

Solution:

$$\begin{aligned} \int_0^1 \mathbf{r}(t) dt &= \left(\int_0^1 \sqrt[3]{t} dt \right) \mathbf{i} + \left(\int_0^1 \frac{1}{t+1} dt \right) \mathbf{j} + \left(\int_0^1 e^{-t} dt \right) \mathbf{k} \\ &= \left[\frac{3}{4} t^{4/3} \right]_0^1 \mathbf{i} + \left[\ln |t+1| \right]_0^1 \mathbf{j} + \left[-e^{-t} \right]_0^1 \mathbf{k} \\ &= \frac{3}{4}\mathbf{i} + \ln 2\mathbf{j} + \left(1 - \frac{1}{e} \right) \mathbf{k}. \end{aligned}$$

5.7 Velocity and Acceleration

We begin by looking at the motion of an object in the plane. As an object moves along a curve in the plane, the coordinates x and y of its center of mass are each functions of time t . Rather than using the letters f and g to represent these two functions, it is convenient to write $x = x(t)$ and $y = y(t)$. So, the position vector $\mathbf{r}(t)$ takes the form

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}.$$

Theorem 5.7.1. If x and y are twice-differentiable functions of t , and $\mathbf{r}$ is a vector-valued function given by $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$, then the velocity vector, acceleration vector and speed at time t are as follows

1. Velocity = $\mathbf{v}(t) = \mathbf{r}'(t) = x'(t)\mathbf{i} + y'(t)\mathbf{j}$.
2. Acceleration = $\mathbf{a}(t) = \mathbf{r}''(t) = x''(t)\mathbf{i} + y''(t)\mathbf{j}$.

3. Speed = $|\mathbf{v}(t)| = |\mathbf{r}'(t)| = \sqrt{[x'(t)]^2 + [y'(t)]^2}$.

Example 5.7.1. Find the velocity vector, speed and acceleration vector of a particle that moves along the plane curve C given by $\mathbf{r}(t) = 2 \sin \frac{t}{2} \mathbf{i} + 2 \cos \frac{t}{2} \mathbf{j}$.

Solution: The velocity vector is

$$\mathbf{v}(t) = \mathbf{r}'(t) = \cos \frac{t}{2} \mathbf{i} - \sin \frac{t}{2} \mathbf{j}.$$

The speed (at any time) is

$$|\mathbf{r}'(t)| = \sqrt{\cos^2 \frac{t}{2} + \sin^2 \frac{t}{2}} = 1.$$

The acceleration vector is

$$\mathbf{a}(t) = \mathbf{r}''(t) = -\frac{1}{2} \sin \frac{t}{2} \mathbf{i} - \frac{1}{2} \cos \frac{t}{2} \mathbf{j}.$$

So far, we have concentrated on finding the velocity and acceleration by differentiating the position vector. Many practical applications involve the reverse problem, finding the position function for a given velocity or acceleration.

Example 5.7.2. An object starts from rest at the point $P = (1, 2, 0)$ and moves with an acceleration of $\mathbf{a}(t) = \mathbf{j} + 2\mathbf{k}$ where $|\mathbf{a}(t)|$ is measured in meters per second per second. Find the location of the object after $t = 2$ seconds.

Solution: From the description of the object's motion, we can deduce the following initial conditions. Because the object starts from rest, we have $\mathbf{v}(0) = 0$. Moreover, because the object starts at the point $(x, y, z) = (1, 2, 0)$, we have

$$\mathbf{r}(0) = x(0)\mathbf{i} + y(0)\mathbf{j} + z(0)\mathbf{k} = 1\mathbf{i} + 2\mathbf{j} + 0\mathbf{k} = \mathbf{i} + 2\mathbf{j}.$$

To find the position function, we should integrate twice, each time using one of the initial conditions to solve for the constant of integration. The velocity vector is

$$\mathbf{v}(t) = \int \mathbf{a}(t) dt = \int (\mathbf{j} + 2\mathbf{k}) dt = t\mathbf{j} + 2t\mathbf{k} + \mathbf{C}$$

where $\mathbf{C} = C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k}$. Letting $t = 0$ and applying the initial condition $\mathbf{v}(0) = 0$, we obtain

$$\mathbf{v}(0) = C_1\mathbf{i} + C_2\mathbf{j} + C_3\mathbf{k} = 0 \implies C_1 = C_2 = C_3 = 0.$$

Thus, the velocity at any time t is

$$\mathbf{v}(t) = t\mathbf{j} + 2t\mathbf{k}.$$

Integrating once more produces

$$\mathbf{r}(t) = \int \mathbf{v}(t) dt = \int (t\mathbf{j} + 2t\mathbf{k}) dt = \frac{t^2}{2}\mathbf{j} + t^2\mathbf{k} + \mathbf{C}$$

where $\mathbf{C} = C_4\mathbf{i} + C_5\mathbf{j} + C_6\mathbf{k}$. Letting $t = 0$ and applying the initial condition $\mathbf{r}(0) = \mathbf{i} + 2\mathbf{j}$, we have

$$\mathbf{r}(0) = C_4\mathbf{i} + C_5\mathbf{j} + C_6\mathbf{k} = \mathbf{i} + 2\mathbf{j} \implies C_4 = 1, C_5 = 2, C_6 = 0.$$

Thus, the position vector is

$$\mathbf{r}(t) = \mathbf{i} + \left(\frac{t^2}{2} + 2\right)\mathbf{j} + t^2\mathbf{k}.$$

The location of the object after 2 seconds is given by $\mathbf{r}(2) = \mathbf{i} + 4\mathbf{j} + 4\mathbf{k}$.

5.8 Tangent Vectors and Normal Vectors

In the previous section, we learned that the velocity vector points in the direction of motion.

5.8.1 Definition of Unit Tangent Vector

Let C be a smooth curve represented by $\mathbf{r}$ on an open interval I . The **unit tangent vector** $\mathbf{T}(t)$ at t is defined to be

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}, \quad \mathbf{r}'(t) \neq \mathbf{0}.$$

Example 5.8.1. Find the unit tangent vector to the curve given by $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j}$ when $t = 1$.

Solution: The derivative of $\mathbf{r}(t)$ is $\mathbf{r}'(t) = \mathbf{i} + 2t\mathbf{j}$. Thus, the unit tangent vector is

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} = \frac{1}{\sqrt{1+4t^2}}(\mathbf{i} + 2t\mathbf{j}).$$

When $t = 1$, the unit tangent vector is

$$\mathbf{T}(1) = \frac{1}{\sqrt{5}}(\mathbf{i} + 2\mathbf{j}).$$

The **tangent line to a curve** at a point is the line passing through the point and parallel to the unit tangent vector.

Example 5.8.2. Find $\mathbf{T}(t)$ and then a set of parametric equations for the tangent line to the helix given by $\mathbf{r}(t) = 2 \cos t \mathbf{i} + 2 \sin t \mathbf{j} + t \mathbf{k}$ at the point corresponding to $t = \frac{\pi}{4}$.

Solution: The derivative of $\mathbf{r}(t)$ is $\mathbf{r}'(t) = -2 \sin t \mathbf{i} + 2 \cos t \mathbf{j} + \mathbf{k}$, which implies that $|\mathbf{r}'(t)| = \sqrt{4 \sin^2 t + 4 \cos^2 t + 1} = \sqrt{5}$. Therefore, the unit tangent vector is

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} = \frac{1}{\sqrt{5}}(-2 \sin t \mathbf{i} + 2 \cos t \mathbf{j} + \mathbf{k}).$$

When $t = \frac{\pi}{4}$, the unit tangent vector is

$$\mathbf{T}\left(\frac{\pi}{4}\right) = \frac{1}{\sqrt{5}}\left(-2\frac{\sqrt{2}}{2}\mathbf{i} + 2\frac{\sqrt{2}}{2}\mathbf{j} + \mathbf{k}\right) = \frac{1}{\sqrt{5}}(-\sqrt{2}\mathbf{i} + \sqrt{2}\mathbf{j} + \mathbf{k}).$$

There are infinitely many vectors that are orthogonal to the tangent vector $\mathbf{T}(t)$.

Definition of Principal Unit Normal Vector

Let C be a smooth curve represented by $\mathbf{r}$ on an interval I . If $\mathbf{T}'(t) \neq 0$, then the **principal unit normal vector** at t is defined to be

$$\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{|\mathbf{T}'(t)|}.$$

Example 5.8.3. Find $\mathbf{N}(t)$ and $\mathbf{N}(1)$ for the curve represented by

$$\mathbf{r}(t) = 3t\mathbf{i} + 2t^2\mathbf{j}.$$

Solution: By differentiating, we obtain

$$\mathbf{r}'(t) = 3\mathbf{i} + 4t\mathbf{j} \quad \text{and} \quad |\mathbf{r}'(t)| = \sqrt{9 + 16t^2}$$

which implies that the unit tangent vector is

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} = \frac{1}{\sqrt{9 + 16t^2}}(3\mathbf{i} + 4t\mathbf{j}).$$

Then differentiating $\mathbf{T}(t)$ with respect to t to obtain

$$\mathbf{T}'(t) = \frac{1}{\sqrt{9 + 16t^2}}(4\mathbf{j}) - \frac{16t}{(9 + 16t^2)^{\frac{3}{2}}}(3\mathbf{i} + 4t\mathbf{j}) = \frac{12}{(9 + 16t^2)^{\frac{3}{2}}}(-4t\mathbf{i} + 3\mathbf{j}).$$

$$|\mathbf{T}'(t)| = 12\sqrt{\frac{9 + 16t^2}{(9 + 16t^2)^3}} = \frac{12}{9 + 16t^2}.$$

Therefore, the principal unit normal vector is $\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{|\mathbf{T}'(t)|} = \frac{1}{\sqrt{9 + 16t^2}}(-4t\mathbf{i} + 3\mathbf{j})$. When $t = 1$, the principal unit normal is $\mathbf{N}(1) = \frac{1}{5}(-4\mathbf{i} + 3\mathbf{j})$.

5.9 Tutorial Questions

1. Find the domain of the vector-valued functions.

(i) $\mathbf{r}(t) = \frac{1}{t+1}\mathbf{i} + \frac{t}{2}\mathbf{j} - 3\mathbf{k}$ (ii) $\mathbf{r}(t) = \sqrt{4 - t^2}\mathbf{i} + t^2\mathbf{j} - 6t\mathbf{k}$ (iii) $\mathbf{r}(t) = \ln t\mathbf{i} - e^t\mathbf{j} - t\mathbf{k}$
(iv) $\mathbf{r}(t) = \sin t\mathbf{i} + 4\cos t\mathbf{j} + t\mathbf{k}$ (v) $\mathbf{r}(t) = \mathbf{F}(t) \times \mathbf{G}(t)$ where $\mathbf{F}(t) = \sin t\mathbf{i} + \cos t\mathbf{j}$ and $\mathbf{G} = \sin t\mathbf{j} + \cos t\mathbf{k}$.

2. Evaluate (if possible) the vector-valued functions at each given value of t .

(a) $\mathbf{r}(t) = \frac{1}{2}t^2\mathbf{i} - (t-1)\mathbf{j}$ at (i) $\mathbf{r}(0)$, (ii) $\mathbf{r}(s+1)$, (iii) $\mathbf{r}(2+\Delta t) - \mathbf{r}(2)$
(b) $\mathbf{r}(t) = \ln t\mathbf{i} + \frac{1}{t}\mathbf{j} + 3t\mathbf{k}$ at (i) $\mathbf{r}(2)$, (ii) $\mathbf{r}(-3)$ (iii) $\mathbf{r}(t-4)$ (iv) $\mathbf{r}(1+\Delta t) - \mathbf{r}(1)$.

3. Find the limit (if it exists).
- (i) $\lim_{t \rightarrow \pi} (t\mathbf{i} + \cos t\mathbf{j} + \sin t\mathbf{k})$ (ii) $\lim_{t \rightarrow 0} \left(t^2\mathbf{i} + 3t\mathbf{j} + \frac{1 - \cos t}{t}\mathbf{k} \right)$
4. Using the definition of the derivative of a vector-valued function, find $\mathbf{r}'(t)$ of the following:
- (i) $\mathbf{r}(t) = 2t\mathbf{i} + \mathbf{j} + \mathbf{k}$ (ii) $\mathbf{r}(t) = 4t\mathbf{i} + 4t\mathbf{j} + 2t\mathbf{k}$ (iii) $\mathbf{r}(t) = t\mathbf{i} + (2t - 5)\mathbf{j} + 3t\mathbf{k}$ (iv) $\mathbf{r}(t) = 3t^2\mathbf{i} + 4t\mathbf{j} - 8t^3\mathbf{k}$.
5. Find $\mathbf{r}'(t)$.
- (i) $\mathbf{r}(t) = (2 \cos t, 5 \sin t)$ (ii) $\mathbf{r}(t) = (t \cos t, -2 \sin t)$ (iii) $\mathbf{r}(t) = (t^3, \cos 3t, \sin 3t)$
 (iv) $\mathbf{r}(t) = 6t\mathbf{i} - 7t^2\mathbf{j} + t^3\mathbf{k}$ (v) $\mathbf{r}(t) = a \cos^3 t\mathbf{i} + a \sin^3 t\mathbf{j} + \mathbf{k}$.
6. Let $\mathbf{r}$ and $\mathbf{u}$ be differentiable vector-valued functions of t and let c be a scalar. Prove the following properties:
- (i) $D_t[\mathbf{r}(t) \pm \mathbf{u}(t)] = \mathbf{r}'(t) \pm \mathbf{u}'(t)$, (ii) $D_t[\mathbf{r}(t) \cdot \mathbf{u}(t)] = \mathbf{r}(t) \cdot \mathbf{u}'(t) + \mathbf{r}'(t) \cdot \mathbf{u}(t)$,
 (iii) If $\mathbf{r}(t) \cdot \mathbf{r}(t) = c$, then $\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$.
7. Find the open interval(s) on which the curve given by the vector-valued function is smooth.
- (i) $\mathbf{r}(t) = \frac{1}{t-1}\mathbf{i} + 3t\mathbf{j}$ (ii) $\mathbf{r}(\theta) = (\theta + \sin \theta)\mathbf{i} + (1 - \cos \theta)\mathbf{j}$ (iii) $\mathbf{r}(t) = \frac{2t}{8+t^3}\mathbf{i} + \frac{2t^2}{8+t^3}\mathbf{j}$
 (iv) $\mathbf{r}(t) = t\mathbf{i} - 3t\mathbf{j} + \tan t\mathbf{k}$ (v) $\mathbf{r}(t) = \sqrt{t}\mathbf{i} + (t^2 - 1)\mathbf{j} + \frac{1}{4}t\mathbf{k}$ (vi) $\mathbf{r}(t) = e^t\mathbf{i} - e^{-t}\mathbf{j} + 3t\mathbf{k}$.
8. Evaluate the indefinite integral.
- (i) $\int (2t\mathbf{i} + \mathbf{j} + \mathbf{k}) dt$ (ii) $\int (3t^2\mathbf{i} + 4t\mathbf{j} - 8t^3\mathbf{k}) dt$ (iii) $\int [(2t - 1)\mathbf{i} + 4t^3\mathbf{j} + 3\sqrt{t}\mathbf{k}] dt$.
9. Find the velocity, speed and acceleration of the object.
- (i) $\mathbf{r}(t) = t\mathbf{i} + (2t - 5)\mathbf{j} + 3t\mathbf{k}$ (ii) $\mathbf{r}(t) = 4t\mathbf{i} + 4t\mathbf{j} + 2t\mathbf{k}$ (iii) $\mathbf{r}(t) = t\mathbf{i} + t\mathbf{j} + \sqrt{9 - t^2}\mathbf{k}$.
10. Consider the motion of a point on the circumference of a rolling circle. As the circle rolls, it generates a cycloid $\mathbf{r}(t) = b(\omega t - \sin \omega t)\mathbf{i} + b(1 - \cos \omega t)\mathbf{j}$, where ω is the constant angular velocity of the circle. Find the velocity and acceleration vectors of the point.
11. Consider a particle moving on a circular path of radius b described by $\mathbf{r}(t) = b \cos \omega t\mathbf{i} + b \sin \omega t\mathbf{j}$, where $\omega = \frac{d\theta}{dt}$ is the constant angular velocity.
- (i) Find the velocity vector and show that it is orthogonal to $\mathbf{r}(t)$.
 (ii) Show that the speed of the particle is $b\omega$.
 (iii) Find the acceleration vector and show that its direction is always towards the center of the circle.
 (iv) Show that the magnitude of the acceleration vector is $\omega^2 b$.